Week #5: Direct Methods for Linear Systems

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Direct methods for solving linear systems $Ax = b$ form the cornerstone of numerical linear algebra. These methods compute the exact solution (in exact arithmetic) through a finite sequence of algebraic operations, typically involving matrix factorizations. This week we develop the mathematical theory underlying these methods, analyzing their stability properties, computational complexity, and practical implementation considerations.

The fundamental approach of direct methods is to transform the original system into an equivalent system that is easier to solve, usually through triangular factorizations. We will explore how different factorization strategies—LU decomposition, Cholesky factorization, and QR decomposition—exploit different matrix properties to achieve computational efficiency and numerical stability.

Our analysis emphasizes the mathematical principles that govern the behavior of these algorithms. We examine how pivoting strategies control numerical stability, how the structure of matrices affects computational complexity, and how error analysis guides the design of robust algorithms. These theoretical insights provide the foundation for understanding when and how to apply direct methods effectively in practice.

1 Gaussian Elimination and LU Decomposition

1.1 Mathematical Foundation

Definition. LU Decomposition

For a matrix $A \in \mathbb{R}^{n\times n}$, an LU decomposition is a factorization $A = LU$ where:

- $L \in \mathbb{R}^{n\times n}$ is lower triangular with unit diagonal ($L_{ii} = 1$)
- $U \in \mathbb{R}^{n\times n}$ is upper triangular

The existence and uniqueness of LU decomposition depends on the properties of the matrix A.

Theorem 1.1 (Existence of LU Decomposition)**.** *Let $A \in \mathbb{R}^{n\times n}$. The LU decomposition $A = LU$ exists and is unique if and only if all leading principal minors of A are nonzero.*

Proof. The proof follows by induction on the matrix dimension, showing that the Gaussian elimination process can be completed without pivoting if and only if no zero pivots are encountered, which occurs precisely when all leading principal minors are nonzero. □

1.2 Gaussian Elimination Algorithm

The Gaussian elimination process systematically eliminates entries below the diagonal through elementary row operations. For a system $Ax = b$, we perform:

$$A^{(0)} = A, \quad b^{(0)} = b \tag{1}$$

$$A^{(k+1)} = M_k A^{(k)}, \quad b^{(k+1)} = M_k b^{(k)} \tag{2}$$

where M_k is the k-th elimination matrix:

$$M_k = I - m_k e_k^T, \quad m_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ m_{k+1,k} \\ \vdots \\ m_{n,k} \end{pmatrix}$$

with multipliers $m_{i,k} = a_{i,k}^{(k)} / a_{k,k}^{(k)}$ for $i > k$.

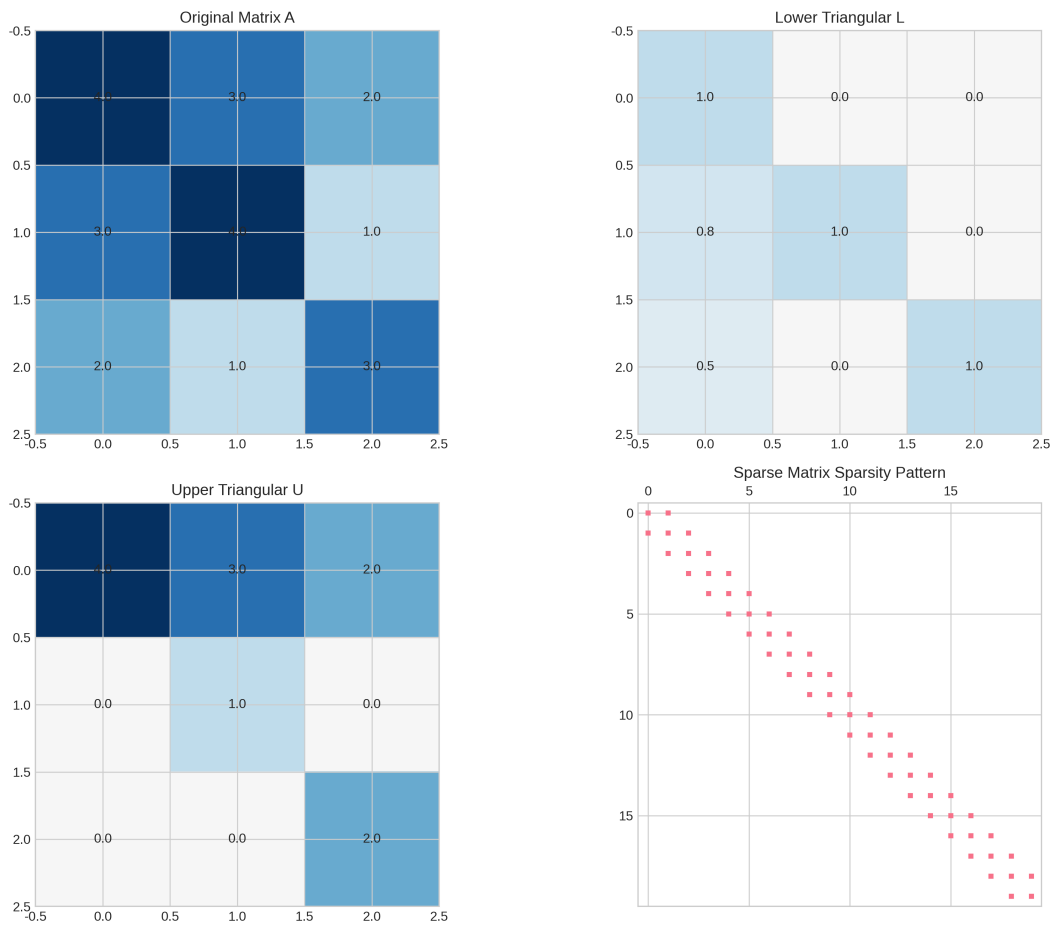

Figure 1: Gaussian elimination process showing the systematic reduction to upper triangular form. The left panel shows the elimination of entries below the diagonal, while the right panel illustrates the corresponding operations on the right-hand side vector.

1.3 Computational Complexity

The operation count for Gaussian elimination is crucial for understanding its practical applicability.

Theorem 1.2 (Computational Complexity of LU Decomposition)**.** *The LU decomposition of an $n \times n$ matrix requires $\frac{2n^3}{3} + O(n^2)$ floating-point operations.*

Proof. The elimination of column k requires approximately $2(n-k)^2$ operations. Summing over all columns:

$$\sum_{k=1}^{n-1} 2(n-k)^2 = 2\sum_{j=1}^{n-1} j^2 = 2 \cdot \frac{(n-1)n(2n-1)}{6} = \frac{2n^3}{3} + O(n^2)$$

□

2 Pivoting Strategies

2.1 Partial Pivoting

Partial pivoting addresses numerical instability by selecting the largest available pivot in each column.

Definition. Partial Pivoting

At step k of Gaussian elimination, choose the pivot row p such that:

$$|a_{p,k}^{(k)}| = \max_{k \le i \le n} |a_{i,k}^{(k)}|$$

and interchange rows k and p.

This leads to the factorization $PA = LU$ where P is a permutation matrix.

Theorem 2.1 (Growth Factor for Partial Pivoting)**.** *With partial pivoting, the growth factor is bounded by 2^{n-1}, but in practice is typically $O(n)$ or $O(n \log n)$.*

2.2 Complete Pivoting

Complete pivoting provides maximum numerical stability at increased computational cost.

Definition. Complete Pivoting

At step k, choose pivot element (p, q) such that:

$$|a_{p,q}^{(k)}| = \max_{k \le i,j \le n} |a_{i,j}^{(k)}|$$

and interchange rows k and p, columns k and q.

This yields $PAQ = LU$ where both P and Q are permutation matrices.

3 Cholesky Decomposition

For symmetric positive definite matrices, the Cholesky decomposition provides a more efficient factorization.

Definition. Cholesky Decomposition

For a symmetric positive definite matrix $A \in \mathbb{R}^{n \times n}$, the Cholesky decomposition is $A = LL^T$ where L is lower triangular with positive diagonal entries.

Theorem 3.1 (Existence and Uniqueness of Cholesky Decomposition)**.** *Every symmetric positive definite matrix has a unique Cholesky decomposition.*

3.1 Cholesky Algorithm

The Cholesky decomposition can be computed using the following recurrence relations:

$$L_{ii} = \sqrt{A_{ii} - \sum_{k=1}^{i-1} L_{ik}^2} \tag{3}$$

$$L_{ji} = \frac{1}{L_{ii}} \left(A_{ji} - \sum_{k=1}^{i-1} L_{jk} L_{ik} \right), \quad j > i \tag{4}$$

Theorem 3.2 (Computational Complexity of Cholesky Decomposition)**.** *The Cholesky decomposition requires $\frac{n^3}{3} + O(n^2)$ operations, exactly half the cost of LU decomposition.*

4 Stability Analysis

4.1 Backward Error Analysis

The stability of direct methods is analyzed through backward error analysis, which determines the perturbation to the original problem that would yield the computed solution as exact.

Theorem 4.1 (Backward Stability of Gaussian Elimination with Partial Pivoting)**.** *Gaussian elimination with partial pivoting is backward stable. The computed solution $\tilde{x}$ satisfies:*

$$(A + \Delta A)\tilde{x} = b$$

where $\|\Delta A\|_\infty \leq \gamma_n \rho \|A\|_\infty$ with $\gamma_n = O(n\epsilon_{mach})$ and ρ is the growth factor.

4.2 Forward Error Analysis

The forward error depends on both the backward error and the condition number of the matrix.

Theorem 4.2 (Forward Error Bound)**.** *For the computed solution $\tilde{x}$ of $Ax = b$, the relative forward error satisfies:*

$$\frac{\|\tilde{x} - x\|}{\|x\|} \leq \gamma_n \rho \kappa(A) + O(\epsilon_{mach}^2)$$

where $\kappa(A) = \|A\|\|A^{-1}\|$ is the condition number.

Intuition: Stability vs. Accuracy

Stability ensures that the algorithm produces a solution that is exact for a slightly perturbed problem. Accuracy depends on both stability and the conditioning of the original problem. A stable algorithm applied to a well-conditioned problem yields an accurate solution.

5 Special Matrix Structures

5.1 Tridiagonal Systems

Tridiagonal systems arise frequently in numerical methods and can be solved efficiently using the Thomas algorithm.

Definition. Tridiagonal Matrix

A tridiagonal matrix has the form:

$$A = \begin{pmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & a_3 & b_3 & \ddots & \\ & & \ddots & \ddots & c_{n-1} \\ & & & a_n & b_n \end{pmatrix}$$

The Thomas algorithm solves tridiagonal systems in $O(n)$ operations through specialized LU decomposition.

5.2 Band Matrices

For matrices with bandwidth $p + q + 1$ (where p is the lower bandwidth and q is the upper bandwidth), the LU decomposition can be computed in $O(npq)$ operations rather than $O(n^3)$.

Theorem 5.1 (Complexity for Band Matrices)**.** *LU decomposition of a band matrix with bandwidths p and q requires approximately npq operations.*

6 Iterative Refinement

Iterative refinement improves the accuracy of computed solutions through residual correction.

Definition. Iterative Refinement Algorithm

Given an approximate solution $x^{(0)}$ to $Ax = b$:

1. Compute residual: $r^{(k)} = b - Ax^{(k)}$
2. Solve correction equation: $A\delta^{(k)} = r^{(k)}$
3. Update solution: $x^{(k+1)} = x^{(k)} + \delta^{(k)}$

Theorem 6.1 (Convergence of Iterative Refinement)**.** *If the LU decomposition is computed with backward error ϵ and $\epsilon\kappa(A) < 1$, then iterative refinement converges and can recover full machine precision accuracy.*

Practice Problems

1. Prove that the multipliers in Gaussian elimination without pivoting satisfy $|m_{ij}| \leq 1$ when partial pivoting is used.
2. Show that for a symmetric positive definite matrix A, the Cholesky decomposition exists and is unique.
3. Analyze the growth factor for the matrix:

$$A = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & 1 & 1 & \cdots & 1 \\ -1 & -1 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & -1 & \cdots & 1 \end{pmatrix}$$

under Gaussian elimination without pivoting.

4. Derive the operation count for solving a tridiagonal system using the Thomas algorithm.

5. Implement iterative refinement and analyze its convergence for increasingly ill-conditioned matrices.

Solution:

Problem 1: With partial pivoting, we always choose the largest available pivot. This ensures that $|a_{kk}^{(k)}| \geq |a_{ik}^{(k)}|$ for all $i \geq k$, which immediately gives $|m_{ik}| = |a_{ik}^{(k)}/a_{kk}^{(k)}| \leq 1$.

Problem 2: For symmetric positive definite A, all leading principal minors are positive (by Sylvester's criterion), ensuring the existence of LU decomposition. Symmetry and positive definiteness guarantee that $L = U^T$, yielding the Cholesky form $A = LL^T$. Uniqueness follows from the uniqueness of LU decomposition when it exists.

Problem 3: This matrix exhibits exponential growth. Without pivoting, the growth factor is exactly 2^{n-1}, demonstrating the worst-case behavior of Gaussian elimination without pivoting.

Problem 4: The Thomas algorithm requires $2n - 1$ multiplications and $n - 1$ divisions in the forward elimination, plus $n - 1$ multiplications and $n - 1$ divisions in the back substitution, totaling $5n - 4$ operations, which is $O(n)$.

Problem 5: Iterative refinement converges when $\epsilon\kappa(A) < 1$. For well-conditioned matrices, convergence is rapid. For ill-conditioned matrices, convergence slows and may eventually stagnate due to the finite precision of residual computation.